

FasalDoc — Executive Summary

AI Crop & Livestock Health Companion for Pakistan Muhammad Zunurain Tahir (Founder & Lead Architect) · Muhammad Abdullah Nadeem (Co-Founder & Product Lead) · Muhammad Abdullah Khalid (Co-Founder & Tech Partner)

The Problem

Smallholder farmers make up roughly 40% of Pakistan's labor force, yet lose an estimated 20-30% of crop yield annually to preventable disease — largely because they have no fast, affordable way to diagnose problems early. Livestock owners face the same gap. Existing global tools like Plantix aren't tuned to Pakistani crops, disease strains, language, or the remedies actually available at a local shop.

The Solution

FasalDoc is a progressive web app that diagnoses crop and livestock diseases from a single phone photo, powered by a multi-provider AI engine (Gemini, Groq, OpenRouter with an agricultural RAG fallback), and returns locally relevant organic and chemical remedies — complete with dosage and local product names — in Urdu or English, working fully offline.

Beyond diagnosis, FasalDoc bundles the tools a farmer actually needs in one place: weather-based disease risk alerts, market price tracking, government scheme discovery, tele-vet booking, and an AI chat assistant for follow-up questions.

Why Now, Why This Team

Smartphone penetration in rural Pakistan has crossed a meaningful threshold, and mobile-optimized AI models have matured enough for a small team to build something genuinely useful without a research lab. No existing product is built for this specific market — and the team combines the technical depth (AI/ML, full-stack, DevOps) to execute across the entire stack.

Traction & Validation Plan

The core diagnosis loop, backend, and mobile experience are built and functioning. The roadmap targets 500 beta farmers in Faisalabad by Q3 2026, a Punjab Agriculture Department partnership and ZTBL integration by Q4 2026, and 50,000 users across 12 crops and 5 livestock categories by Q1 2027.

Business Model

FasalDoc intentionally delays monetization in Year 1 to prioritize adoption and local data collection — the real defensibility moat. Revenue ramps from Year 2 through freemium subscriptions, B2B2C licensing to agri-input companies, and government/NGO program contracts, with marketplace commission added in Year 3. Full projections are detailed in the accompanying Financial Projections workbook.

The Ask

FasalDoc is seeking PKR 2,000,000 in hackathon/seed support, allocated primarily to local data collection and agronomist validation, cloud/API infrastructure, and pilot expansion support — detailed in the Funding Ask tab of the Financial Projections workbook.

This summary accompanies the full Founding Blueprint, Business Model Canvas, Technical Architecture document, and Financial Projections workbook.

